

ACTOR (Schesch and Ernst): CodeWeaver evidence profile

This profile audits every empirical surface identified for ACTOR (Schesch and Ernst) and records the maximum scientifically defensible CodeWeaver comparison. Status: reference_87_and_measured_public_95_overlap. Paper values retain denominator 87, but the pinned results submodule contains 95 project directories and does not identify the paper's 13 exclusions; CodeWeaver's matching public 95 are reported separately.

Evidence boundary

Paper values and CodeWeaver measurements remain separate unless subject identity, revision, denominator, and fixed oracle are verified. Missing values are not converted to zeros.

Existing result

results/crust-bench-codeweaver-comparison-2026-08-14

Complete empirical-surface audit

key	surface	denominator	metrics	status
actor-schesch-ernst	Figure 3	338 TRACTOR cases x 11 systems	compiles, passes, LoC, unsafe percent	exact_reference_public_artifact
actor-schesch-ernst	Figures 4-5	failure classes and 382 output files	failure and unsafe root-cause taxonomy	exact_reference_public_artifact
actor-schesch-ernst	Figure 6	87 CRUST-Bench projects x systems/settings	builds, tests, LoC, unsafe percent	exact_reference_denominator_unresolved_separate_public_95_overlap
actor-schesch-ernst	Figure 7	TRACTOR and two CRUST-Bench modes	prompt-sensitivity ablations	exact_reference_public_artifact
actor-schesch-ernst	Cost analysis	CRUST-Bench, TRACTOR and ablations	USD, minutes and per-kLoC cost	exact_reference_public_artifact

Inclusion and execution decision

paper_id	source_url	empirical_scope	crust_role	codeweaver_status	reason	existing_result
IEEE S&P; 2026	https://homes.cs.washington.edu/~mernst/pubs/c-to-rust-ieee-sp2026-abstract.html	paper: 87 CRUST-Bench projects and 338 TRACTOR cases; public result artifact: 95 CRUST project directories	paper's 87-project exclusion set is not recoverable; separately labeled public-artifact 95-project overlap	reference_87_and_measured_public_95_overlap	paper values retain denominator 87, but the pinned results submodule contains 95 project directories and does not identify the paper's 13 exclusions; CodeWeaver's matching public 95 are reported separately	results/crust-bench-codeweaver-comparison-2026-08-14

actor_schesch_figure6_crust

system	setting	denominator	builds	tests	loc	unsafe_percent
ACTOR Kiro	no benchmark tests	87	82	56	43000	1
C2Rust/Laertes /C2SaferRust/SmartC2Rust	no benchmark tests	87	0	0		
GPT-5.4	no benchmark tests	87	82	50	57000	0
Kimi K2.5	no benchmark tests	87	46	31	28000	0
Gemini 3.1 Pro	no benchmark tests	87	11	8	15000	0
ACTOR Kiro	test repair	87	87	82	52000	1
ACTOR Claude	test repair	87	85	75	45000	0
ACTOR Codex	test repair	87	87	81	48000	1
GPT-5.4	test repair	87	79	64	58000	0
Kimi K2.5	test repair	87	45	39	28000	0
Gemini 3.1 Pro	test repair	87	8	7	15000	0

actor_schesch_figure3_tractor_totals

system	compiles	passes	denominator	loc	unsafe_percent
ACTOR Kiro	338	325	338	47000	50
ACTOR Claude	338	319	338	53000	53
ACTOR Codex	337	244	338	36000	39

system	compiles	passes	denominator	loc	unsafe_percent
ACTOR Kiro no validation	337	230	338	37000	50
C2Rust	205	204	338	87000	70
Laertes	202	201	338	88000	66
C2SaferRust	193	154	338	82000	59
SmartC2Rust	48	40	338	7000	2
Kimi K2.5	157	118	338	25000	17
GPT-5.4	189	154	338	28000	10
Gemini 3.1 Pro	186	156	338	24000	15

actor_schesch_figure4_failures

root_cause	count
undefined behavior	3
macros	3
configuration	1
input processing	9
underspecified	2
truncated output	12

actor_schesch_figure5_unsafe

root_cause	count
C string/pointer conversion	2194
raw-pointer signatures/casts	5957
pointer arithmetic	3931
C ABI preservation	1648
ptr read/write/copy	527
FFI calls	208
mutable global	116
uninitialized structs	70
bridging raw pointers	60
function-pointer dispatch	36
other	44

actor_schesch_figure7_prompts

configuration	tractor_passes	tractor_denominator	crust_no_tests_passes	crust_test_repair_passes	crust_denominator
ACTOR Claude	319	338	56	75	87
without subtask	313	338	56	79	87
without iteration	249	338	31	73	87
without features	204	338	55	76	87
minimal	171	338	41	75	87

actor_schesch_cost

system	scope	cost_usd	minutes	cost_per_kloc_usd	minutes_per_kloc
ACTOR Kiro	CRUST-Bench	67		1.57	19
ACTOR Kiro	TRACTOR	93		1.97	34
ACTOR Kiro	largest P01 case	3.61	76		
Claude Code	TRACTOR	570			
all configurations	full ablation	2900			

CodeWeaver public-artifact 95-project overlap

system	public_artifact_projects	paper_projects	paper_membership_status	build_successes	test_successes	fixed_tests_passed	fixed_tests_expected
CodeWeaver repetition 1	95	87	unresolved; not treated as the paper's 87-project slice	95	56	425	564
CodeWeaver repetition 2	95	87	unresolved; not treated as the paper's 87-project slice	95	55	433	564
CodeWeaver repetition 3	95	87	unresolved; not treated as the paper's 87-project slice	95	54	403	564